

Independent machine verification of the fundamental-group computation in Wuebben’s exotic $S^2 \times S^2$

John Clyde

VentiMath

`verification@ventimath.org`

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Abstract

Wuebben (arXiv:2608.17267), building on Lidman–Piccirillo (arXiv:2505.14387), proves that a specified Lidman–Piccirillo piece V — a symplectic 4-manifold with the homology of $S^2 \times D^2$, built from a non-product genus-2 surface bundle over a once-punctured torus by two Luttinger surgeries — is simply connected; the symplectic double of V is then homeomorphic but not diffeomorphic to $S^2 \times S^2$, and Lidman–Piccirillo regluing yields an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$. The critical step is the simple connectivity: the triviality of the filled fundamental groups for the paper’s permitted surgery parametrizations. We report an independent implementation of that step. The bundle is rebuilt as a simplicial complex from the paper’s marked-fiber data alone; the surgery tori, based meridians, and framed longitudes are extracted combinatorially; and the four theorem-critical filled presentations are proved trivial by complete confluent rewriting, each computation emitting a replayable derivation certificate that two independently written checkers accept. The framing lemma connecting those combinatorial longitudes to the Lagrangian framing is machine-checked with its analytic normalizations rebuilt constructively. Stated carefully: the simple-connectivity proof appears complete relative to explicitly named standard topology theorems, with no known project-specific gap. The theorem is Wuebben’s; this note claims only the verification. All certificates, code, and provenance are public in a pinned release.

1 Introduction

Claimed constructions of small exotic 4-manifolds have repeatedly failed in one place: the fundamental-group computation. The manifolds of [1] rest on a specified Lidman–Piccirillo piece V , obtained by two Luttinger surgeries [5, 3] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a once-punctured torus, following the framework of [2]. Exoticness of the double of V rests on V being simply connected: the filled fundamental groups of the paper’s permitted surgery parametrizations — the four sign choices of the double surgery in the paper’s principal coordinate system — must be trivial.

This note reports an independent implementation of that step, together with a machine audit of the framing lemma that connects the group theory to the symplectic geometry. Nothing in the implementation reuses the paper’s coordinates, code, or intermediate data: the geometry was rebuilt from the paper’s stated marked-fiber data alone, and the paper’s displayed relations, sign tables, and Table 1 filling words were then *compared against* the

rebuild rather than assumed. The downstream deductions — from simple connectivity to the homeomorphism classifications and the exoticness conclusions — are Wuebben’s; this note verifies the simple-connectivity input and its geometric framing bridge.

The result separates into an unconditional computational theorem and an identification that connects it to the paper.

Theorem 1 (certified triviality). *Let $P_{\epsilon_a \epsilon_b}$, for $(\epsilon_a, \epsilon_b) \in \{\pm\}^2$, be the four filled presentations of Section 4: the exported 4-generator, 95-relator presentation of the complement of the two surgery tori in the simplicial model, together with the two filling relators of the $n=0$ double surgery with signs (ϵ_a, ϵ_b) , imposed with the certified based meridians and framed longitudes. Each $P_{\epsilon_a \epsilon_b}$ presents the trivial group. The proof is a derivation certificate, contained in release v1.4 of the verification repository, that replays from a fresh checkout and is accepted by two independently written checkers.*

Lemma 2 (certificate soundness). *A certificate is a finite sequence of records, each an input relator, an inverse axiom of the free group, a critical-pair overlap of two earlier records, or a rewrite of an earlier record by earlier records, ending in the rules $g \rightarrow 1$ for every generator g . A checker accepts only if every record is derived as stated from records preceding it. Every accepted record is then an equality holding in the group presented by the input relators, so acceptance implies that this group is trivial. Neither the confluence of the completed rewriting system nor the correctness of the program that produced it enters the argument.*

Proposition 3 (identification). *The presentations of Theorem 1 present $\pi_1(V)$ for the paper’s four permitted parametrizations: the simplicial model is identified with the paper’s marked bundle by a constructed fiberwise homeomorphism (Section 2); its peripheral loops are based meridians and product-framed longitudes of the surgery tori (Section 3); and the product framing agrees with the Lagrangian framing of Luttinger surgery by the paper’s Lemma 8.2, machine-checked (Section 3).*

Theorem 1 is unconditional: it is a finite object that any reader can re-verify. Proposition 3 is where the residual risk of any machine verification lives, since a machine certifies properties of a model; it rests on the reading of the paper’s conventions, on elementary inputs of PL topology, and on the named theorems of Section 6. The accurate summary of its status is: *the simple-connectivity proof appears complete relative to explicitly named standard topology theorems, with no known project-specific gap* — a statement about verification depth, offered to facilitate expert review, not to replace it.

Two disclosures. The theorem is Wuebben’s; this note asserts nothing about priority or authorship of the mathematics and claims only the verification. The verification was carried out by generative-AI systems under the author’s direction, with commit-level provenance recorded in the repository’s ledger and the corrections produced by adversarial cross-model review preserved rather than rewritten; mathematical responsibility rests with the human author. No claim below depends on trusting a model: every asserted computation ships with a replayable certificate.

2 The independent model and the interpretation dictionary

The genus-2 fiber is realized as a regular neighborhood of the paper’s five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0

realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α , T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

The identification with the paper’s marked bundle is by literal based monodromy action: the based mapping classes of both monodromies are certified as equalities of vertex paths, transported by proof-producing Tietze chains that replay step by step, and completed by an explicit marked mapping-cylinder clutching over the base graph. The identification is thus a constructed fiberwise homeomorphism into the paper’s bundle rather than an appeal to the classification of surface bundles. Everything the surgery argument needs is transported through that map into the paper’s already-smooth manifold, where the smooth and symplectic arguments run; no smoothing of the triangulation is ever chosen.

Because the machine certifies properties of a model, the model itself is the residual risk. Five structurally different implementations reconstruct it independently, arranged so that each class of translation error — subdivision artifacts, monodromy transcription, ordering, sign, and orientation conventions — would be caught by at least one of them. What redundancy cannot exclude, a shared misreading of the paper’s conventions, is addressed by a bound interpretation dictionary: every convention read from the paper carries the certified computation that would fail under a rival reading. Wuebben’s committed scripts enter at exactly one point, parsed and never executed: after the dictionary was frozen, selected entries were compared diagnostically against them, and no verified conclusion consumes script-derived data. The remaining transcription-level entries are audited by a quarantined extractor against the hash-pinned immutable Lidman–Piccirillo v1 arXiv source [2], in both its prose and its vector-figure layers; the one interpretation remaining at that boundary is the ordinary convention that dashed arcs in a handle picture depict hidden projection rather than extra crossings. The full inventory of implementations and the dictionary are in the repository.

3 Peripheral data and the framing identification

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper’s own basing whiskers. The paper’s sign-correction package (its §7.4 fix) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper’s corrected convention requires. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked.

Luttinger surgery is performed with respect to the Lagrangian framing; the paper’s Lemma 8.2 identifies it with the fibered framing carried by the combinatorial longitudes above. This lemma carries an exact logical load: if the identification failed by j meridians, the certificates of Theorem 1 would remain correct for the product-framed presentations they literally describe, but would no longer identify those presentations with the paper’s surgeries. The lemma’s displayed Weinstein-chart, seam, and constant-momentum algebra are machine-checked symbolically, and its inline analytic normalizations are rebuilt constructively rather than cited: the relative Moser flow is produced as the explicit monotone inverse of a cumulative coordinate, and the equivariant step defines the Moser field directly on the double

cover with certified projection and deck invariance, so neither an ODE existence theorem nor a covering-lift theorem is invoked. Chart independence is proved between the paper’s displayed charts by two routes, with the one symplectic move that could introduce meridian components — the closed-1-form shift — excluded by the zero-section condition; [3] (§2.1 and Proposition 2.2) stands as corroboration.

Independently of the lemma, a finite scan tests what a framing error would do. It refills the complement with longitudes shifted by (j_a, j_b) meridians on (T_α, T_β) for the eight axis shifts $(j_a, 0)$, $(0, j_b)$ with $0 < |j| \leq 2$ and the four diagonal shifts $(\pm 1, \pm 1)$ — a cross-and-diagonal sample, not the full grid — across all four sign pairs and both half-drift families, giving 96 counterfactual presentations, plus four zero-shift controls that re-derive the paper’s own $n=0$ fillings through the same pipeline. All 100 present the trivial group, and no nontriviality witness was found at any stage. The evidence is of two levels. The scan pipeline, which simplifies each presentation before attempting completion, decided 82 cases at the level of a GAP session — 40 by Tietze collapse of the simplified presentation and 42 by its confluent completion — and those runs exported no certificates; the four zero-shift controls among them are the paper’s own fillings, whose separate certificates are those of Theorem 1. The 18 it could not decide, every one involving a shift on T_α , carry derivation certificates of the kind in Section 5: 17 from confluent completion of the unsimplified 97-relator presentation, whose redundant relators are what make completion tractable, and the last, the $n=0$ $(+, +)$ filling shifted by $(+1, +1)$, from a certified reduction through the 96 relators it shares with its $n=1$ companion, which complete confluent to an infinite cyclic group in which the omitted relator reduces to a generator. Within the scanned range, then, no framing error would have produced a nontrivial group. The scan is robustness evidence only: the paper’s own fillings are the certified controls, and it bears on the proof only if Lemma 8.2 first fails. The table, methods, and search history are in the repository.

4 Certified triviality of the four theorem-critical groups

Filling the triangulation-derived complement presentation (4 generators, 95 relators) with the certified coherent peripheral pairs yields, for all four $n=0$ sign pairs, complete confluent rewriting systems in which every generator reduces to the identity. Each completion is exported as a derivation certificate, and Theorem 1 is the statement that the four certificates verify.

system	signs	derivation records
$n=0$	$(-, -)$	1123
$n=0$	$(-, +)$	1075
$n=0$	$(+, -)$	804
$n=0$	$(+, +)$	1504

Derivation-DAG certificates for the four theorem-critical filled groups; each verifies TRIVIAL from a fresh clone of the repository.

Four adjacent $n=1$ half-drift presentations of the same fillings — re-indexings that carry no independent logical load — pass the identical certification as an auxiliary check; their certificates sit beside these in the same manifest.

5 Certificate verification

Knuth–Bendix completion (KB MAG [11], run under GAP [12]) is used only as an *untrusted certificate generator*. A patched build of KB MAG records the ancestry of every equation it derives; an untrusted compiler prunes that history to the dependency cone of the final identity rules and emits a certificate of the form in Lemma 2, binding the input presentation by SHA-256. A small independent checker then re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, tidying change, and final generator-to-identity rule; it neither imports nor runs KB MAG. A second checker, written independently in Ruby, re-verifies all retained derivation records against the same hash-bound certificates and accepts, importing neither the first checker nor any project group code. All certificates, the exported rewriting systems, and a hash manifest replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KB MAG independently of the certificate path. The design principle throughout is that certificates are checked, not trusted: search and completion tools are untrusted generators, and every computational claim rests on a replayable artifact validated by small independent code.

6 Trust boundary and data availability

Inputs to the verification result

Theorem 1 has no mathematical input beyond Lemma 2; its trust rests on the two small checkers. Proposition 3 rests on elementary inputs of PL topology retained as standard without a specific citation — the ribbon-thickening and bistellar-trace interpretations that read the paper’s figures into complexes, the classification of surfaces, the collapsible-3-ball and cyclic-knot-unknot criteria behind the local-flatness certificates, and simplicial fundamental-group presentation with Tietze theory — together with the Kerékjártó periodic-disk involution theorem [4] behind the disk extension of the paper’s Lemma 7.1, and [3] as corroboration of the framing identification. The ribbon-interpretation input is narrower than the phrase suggests: the certified classification of equivariant ribbon rotations pins the implication from the paper’s stated marked five-chain data to the equivariant normal form, and the stated data themselves are audited in the original Lidman–Piccirillo source, so what this boundary retains is the thickening of that data into complexes together with the dashed-arc convention of Section 2. This note reproves none of these.

Downstream theorems audited, not depended on

The deductions from simple connectivity to exoticness are Wuebben’s and are inputs to nothing above. Their named inputs — Freedman’s simply connected classification [6] with the Hambleton–Kreck finite-cyclic classification [7, 8]; symplectic Kodaira dimension and minimality via the Luttinger-surgery constructions [3, 9]; the symplectic Thom theorem [10]; and the Rokhlin/Arf input of the slicing argument together with the Ozsváth–Szabó / Lidman–Piccirillo Floer machinery [2] — were audited against their hypotheses, and no gap was found. That audit is recorded in the repository and is not part of this note’s claim.

Data availability

All permanent certificates, frozen inputs, replay scripts, hash manifests, run transcripts, and provenance records are contained, with clean-checkout replay instructions, in the tagged release v1.4 of the repository: <https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v1.4>. The `verification/` tree of that release is identical to that of the repository commit

3f269d3ccee84d04b3e2bfcc1c5875f55a5112f

and its filled-group proof manifest has SHA-256

c79ab0fe97049a7e437f540b274d6425d0bfae2d5a53753e2c349c5351f92da5

The raw completion histories from which certificates are compiled are reproducible build products and are omitted. The paper’s own code is at <https://github.com/bwuebben/exotic-s2xs2>; nothing in the verification derives from it except where explicitly cited.

References

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